

Anfa Majid

✉ anfamajid@hotmail.com 📞 (+92) 333-2404665 🌐 anfamajid 🌐 anfa-majid

EDUCATION

Institute of Business Administration (IBA), Karachi
BS Computer Science

2021 – 2025
CGPA: 3.37/4.0

RESEARCH EXPERIENCE

When Equal Forecast Errors Lead to Unequal Outcomes: Error Semantics in Kubernetes Autoscaling
Independent Research – Research Manuscript

2026
GitHub

- Investigated why forecasts with matched **MAE and RMSE** can produce different Kubernetes autoscaling outcomes due to forecast error semantics, including error direction, timing, event structure, transition alignment, shape, and displacement.
- Designed and implemented a reproducible **three-node K3s experimental framework** with a deterministic predictive autoscaler, CPU-controlled HTTP workload, controlled forecast traces, and instrumentation spanning replica decisions, Deployment state, Pod readiness, service endpoints, and request-level performance.
- Conducted controlled experiments using **14 forecast traces organized into seven matched-error pairs**, with eight repetition blocks per pair, isolating semantic differences while holding aggregate forecast accuracy approximately constant.
- Demonstrated that accuracy-matched forecast errors can diverge operationally: on a narrow-spike workload, a missed peak produced **180 additional deficient command-replica-seconds** and substantially higher request P99 and SLO-violation duration than an equally sized false peak.

EXPERIENCE

Associate Software Development Engineer – Veeam
Karachi, Pakistan

June 2025 – Present

- Replaced a kernel-mount SMB connector with a custom **gosmb2** fork supporting Kerberos and NTLM authentication, doubling L1 scan throughput from 3M to **6M files/hour** and reducing per-million-file inventory time from **40 to 21 minutes**.
- Designed and implemented **stateless backend services** for connectors on AWS Outposts, enabling ephemeral per-scan Kubernetes pods and horizontal scaling across HDFS, NetApp, Azure Blob, NFS, and SMB storage backends.
- Engineered production-grade IaaS connectors for **AWS, Azure, GCP, and OCI** in Golang, enabling automated data compliance scanning across heterogeneous multi-cloud enterprise environments.
- Built the NetApp entitlement data layer to extract **file-level ACLs, user/group permissions, appliance groups, and Active Directory memberships**, and developed a Databricks pipeline to normalize entitlement data into a unified access inventory.
- Consolidated per-connector Docker images into a shared base image with **runtime dispatch**, reducing redundant appliance loads and improving deployment efficiency across storage backends.

Backend Intern – Securiti.ai
Karachi, Pakistan

Jun 2024 – Aug 2024

- Prototyped an AI-powered marketing review tool with video preprocessing, model serving, and MinIO storage; containerized the system using Docker.

Software Engineering Intern – Bitnine Global
Remote

Mar 2024 – May 2024

- Contributed to **Apache AGE**, a PostgreSQL graph extension, through bug fixes, documentation, and feature development in collaboration with core maintainers.

SELECTED PROJECTS

NeuralTrace – Multimodal AI for Memory Assistance

- Developed a **FastAPI/PostgreSQL backend** supporting asynchronous ML inference, semantic vector retrieval, caregiver geofencing, and real-time alerts for a memory-assistance application.
- Implemented a multimodal perception pipeline combining **YOLOv8 and ResNet-50**, achieving 84% indoor scene classification accuracy through feature fusion and co-occurrence analysis.
- Implemented privacy-preserving semantic logging and FAISS-backed vector retrieval, achieving **sub-200ms retrieval latency** and 92.2% transcription accuracy.

Generic Multi-Source Lakehouse Pipeline

- Designed a metadata-driven **Databricks Lakehouse** pipeline ingesting batch and streaming data from S3, SMB, and Excel using Medallion Architecture, incremental scans, and schema auto-discovery.
- Replaced source-specific hardcoding with a runtime metadata store supporting type casting, normalization, and canonical schema mapping across heterogeneous data sources.

TECHNICAL SKILLS

- **Languages:** Go, Python, Java, SQL
- **Systems:** Kubernetes, Docker, Linux, REST APIs, Distributed Systems
- **Cloud:** AWS, Azure, GCP, OCI
- **Data:** PostgreSQL, MongoDB, Kafka, Hadoop, Spark, Databricks, Airflow
- **Storage:** S3, SMB, NFS, HDFS, NetApp, MinIO

LEADERSHIP

- Executive Assistant, IBA Karachi; Advisor, IBA Data Science Society
- Head of Department, IBA Computer Science Society & IBA Arts and Photography Society
- Client Manager, IBA Public Speaking Society